Sardar Patel Institute of Technology

Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai: 400058, India
(Autonomous College of Affiliated to University of Mumbai)

End Semester Examination
December 2023

Maxi Marks: 100 Duration: 3 hours
Class: T.E Semester: V
Course code: CS301 Branch: COMP/CSE-AIML/CSE-DS
Name of the course: Theory of Computation

Instructions:
(1) All questions are compulsory
(2) Partial credit will be given, so show your work where appropriate.
(3) Try to be precise in your answers in order to maximize your points.
(4) Note: DFA = Deterministic Finite Automata, NFA = Nondeterministic Finite Automata, PDA = Push-Down Automata, CFG=Context Free Grammar, TM=Turing Machine.
(5) Let wR denote the reverse of a word w. That is, if w =a1a2 · · · an, then wR = an · · · a1.

Q No		Max Marks	CO	BL
Q1 A	Construct a ε-NFA for regular expression (11 + 10) * and convert it into deterministic finite automata.	8	1	3
B	Construct the deterministic finite automata that accepts set of strings over the alphabet Σ = {a,b}containing either "aab" or "bba".	7	1	3
C	Give the formal definition of Moore machine. **OR** Construct the ε- NFA for the following regular expressions: 1. RE = (a*b + b*a)* 2. RE = (0+1)* (00+11)	5	1	3
Q2 A	Design a mealy machine for RE = (0 + 1)* (00 + 11) and convert it into Moore machine.	8	1	3
B	Simplify the following grammar. S → aS | bA A → bA| B| ε B → a | ε	7	3	3
C	Using pumping lemma prove that following language is not regular. L = $\{a^n b a^{3n} \mid n \geq o\}$	5	2	3
Q3 A	Construct the regular expressions for the following. 1. Set of string defined over the alphabet Σ = {a,b} divisible by 3. 2. L = {W | W has a '1' at every odd position and |W| is odd for Σ = {a,b}.} 3. L = {UVW | U, W belongs to Σ* and |V| = 2 where Σ = {a,b} .}	6	2	3
B	Let G be the grammar. Find the Leftmost and rightmost derivation for the string 001222. G: S → 0S | 1A | 2B | ε A → 1A| 2B| ε B → 2B | ε	4	3	3

Q4 A	Convert the following grammar into Chomsky Normal form (S is the start variable): $S \rightarrow AAA \mid b$ $A \rightarrow Aa \mid B$ $B \rightarrow \varepsilon$	8	3	3
B	Give equivalent PDA for the following grammar (S is the start variable): $S \rightarrow aAA$ $A \rightarrow aS \mid bS \mid a$	5	4	3
C	Give Turing machines (with informal as well as formal descriptions) for the following language: Given a word 'w' on the first tape, copy the word on the second tape and make the first tape completely blank.	7	4	3
Q5 A	Prove or disprove the following statement: Let L and L' be languages such that L is reducible to L' and L is not Turing-recognizable then L' is also not Turing-recognizable.	5	4	1
B	Prove or disprove the following statement: There is a language L such that L is undecidable and $L \leq \overline{L}$	5	4	4
Q6 A	Let <M> denote the description of a Turing machine M. Let L1 = {<M> | M accepts input <M>}. Is L1 decidable? Is L1 recursively enumerable? Justify your answers.	8	4	4
B	Construct a CFG which can generate the following language: L = {w ε {a,b}* | w has at least 3 a's} **OR** Give an algorithm to decide whether the language generated by a given CFG is finite.	7	3	3
C	State True or False with justification. Let L1 and L2 be two languages such that L1 polynomial time reduces to L2. If L2 belongs to the class P, then L1 also belongs to the class P.	5	4	2